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Research question

The specific question of this paper is “What is the relationship between the surface area of a magnet and the time it floats on top of a superconductor due to flux pinning?”. In other words, I will investigate to what extent the surface area of a magnet depends on the flux tube strength to stay pinned. Are these values proportional?

Introduction

The process of flux pinning has many helpful uses, specifically in the space world. Research on this concept can lead to this process being used in future spacecraft. Space makes Flux Pinning possible because of its extremely cold temperatures. The temperatures under -270.45 degrees Celsius provide ideal conditions for this process as many materials have critical temperatures that are adjacent to this temperature and allow for superconductors to activate. Furthermore, the cold temperature of space prevents the superconductor from rising in temperature and losing its superconducting properties.

Contrary to this, the earth does not have these ideal temperatures which makes Flux creep, the slowing of flux motion at temperatures change, a limitation for the study within this paper. This phenomenon makes it so that the pinning potential decreases with increasing temperature. Space’s low temperatures allow for superconductor’s currents to continue being excited and therefore allow spacecraft to connect with another spacecraft without being physically attached. This process could help with further space research, allowing agencies like NASA to no longer have to develop an actuated mechanism to connect space crafts and therefore help Earth further its space research by saving fuel and traveling further (Peck et al.).

The goal of this work is to understand and explore the patterns that can be seen in the process of flux pinning.

Background: The process of Flux pinning

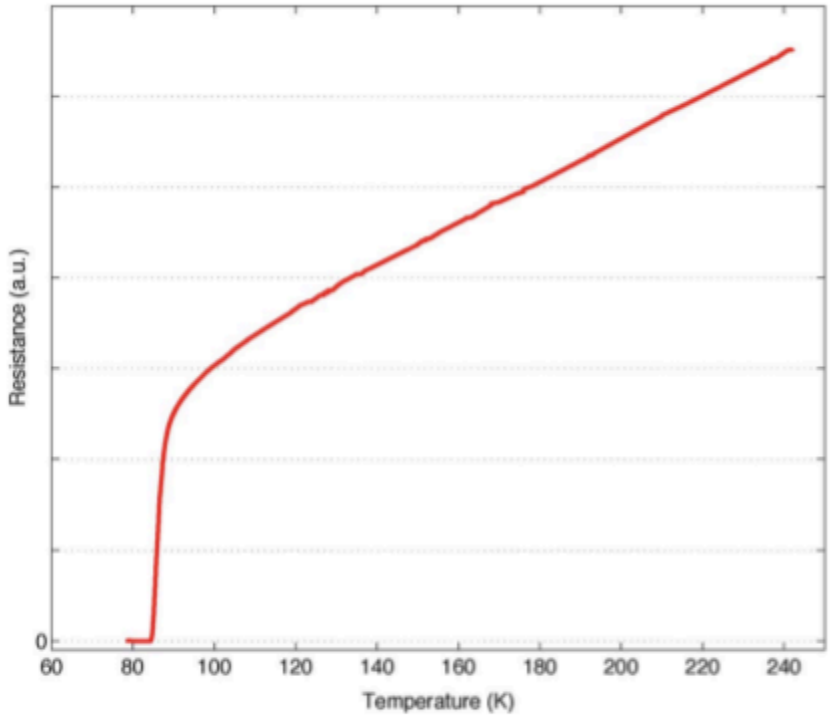


Figure 1: Resistance vs temperature graph of YBCO (Ghamsari)

Quantum locking –also referred to as flux pinning– is a process that consists of a superconductor "locking" in a position due to being enclosed by a magnetic field.

Superconduction: Superconductors include materials that can reach zero resistance. This is done at optimal temperatures referred to as Critical Temperatures. Each superconducting material has a different critical temperature. This exploration will work with a Type II YBCO superconductor.

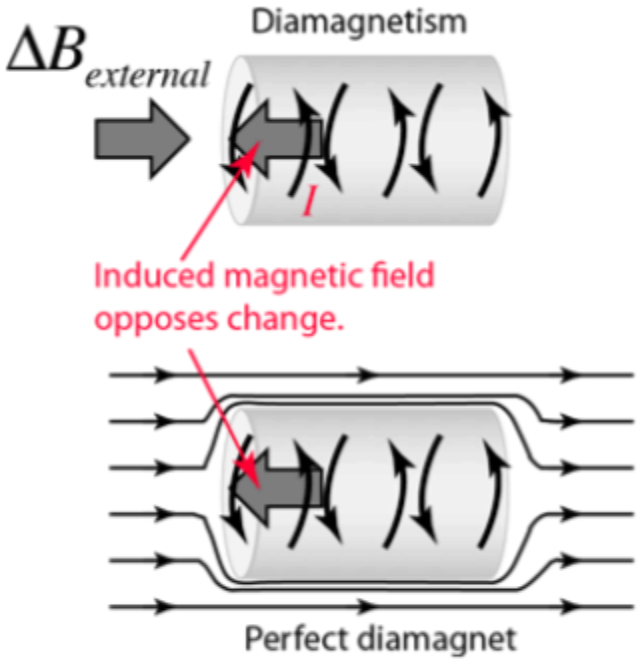


Figure 2: Diagram representing the direction of the magnetic field within the superconductor ("Meissner effect for superconductors").

The process of Quantum locking: Liquid nitrogen's temperature of 77 kelvin allows for the superconductor used within this study - Yttrium Barium Copper Oxide– to reach an optimal temperature of 90 kelvins. As previously mentioned, approaching the critical temperature results in there being no internal resistance within the superconductor. According to the *Bardeen, Cooper, and Schrieffer Theory,* this means that the superconductor's crystal lattice has negligible vibrations – kinetic energy– which facilitates the creation of bosons named Cooper pairs (Wogan). These Cooper particles move through the superconductor and don't bump into the atoms which creates a "phase-coherent condensate that can flow through a material without scattering" (Wogan). Furthermore, impurities within the superconductor such as areas in the material that vary in thickness act as weak points. These points facilitate the process of the Magnet's magnetic field to get through and overcome the Meissner Effect. The Meissner effect refers to the expulsion of the magnetic field from the superconductor as it approaches its critical temperature. This causes a back-and-forth attraction and repulsion. When a magnet is put near an activated superconductor, this phenomenon gives the superconductor the ability to "perfectly match the magnetic fields surrounding it" (Jan et al.) achieving equilibrium and therefore balancing the superconductor.

Hypothesis

As time progresses the superconductor wants to achieve thermal equilibrium with its environment, which results in it reaching a temperature that surpasses its critical temperature and therefore stops its superconducting properties. By using Newtown's law of cooling I can find the time needed to ensure that the superconductor reaches its critical temperature (Feldman et al.). I can also use this to get an expected value for the time it would take for the superconductor to go from its superconducting state back to its normal state. This estimated time would be the maximum amount of time that the superconductor could be suspended due to quantum locking. Furthermore, the time found through this equation will help evaluate whether the findings I had were significant or if there were too many factors that I didn't consider.

Newtons Law of Cooling:

$$T(t) = T_s - (T_s - T_0)e^{-k}$$

Newton's Law of Cooling solved for t (time):

$$\frac{ln\left(\frac{T_s - T(t)}{T_s - T_0}\right)}{-k} = t$$

Sample calculation:

$$\frac{ln\left(\frac{300-90}{300-77}\right)}{-4.6661\times10^{-3}} = 12.874$$

Specific heat capacity (k) of YBCO: 4.6661 ×10 -3 eV/K (7.4658 ×10 -22 JK -1)
Room temperature: $T_s = 300k$
Initial temperature $T_0 = 77k$
Final temperature $T(t) = 90k$
Time required to alternate between critical and room temperature $t = 12.874$

With the surrounding environment's temperature being taken into account, I can evaluate the time that my superconductor quantum locked– with differently sized magnets– compared to this value. Because all of the magnets I used were made of the same material, as the surface area of the magnet increases, there will be a stronger magnetic field (Quantumlevitation.com).

Furthermore, the bigger magnets will allow the superconductor to levitate longer because their area is proportional to a stronger magnetic field. This is due to the fact that magnetic penetration is what makes Quantum locking possible. I hypothesize a linear pattern between the area of the magnets used, and the time the superconductors levitation time because of the direct relationship between magnetic flux and magnetic field.

$$\Phi = BA\cos\theta$$

Φ = magnetic flux

B = magnetic field

A = area

θ = angle between a perpendicular vector to the area and the magnetic field

Figure 3: Formula for magnetic flux ("22.1: Magnetic Flux, Induction, and Faraday's Law")

Set-Up

Proposed Procedure:

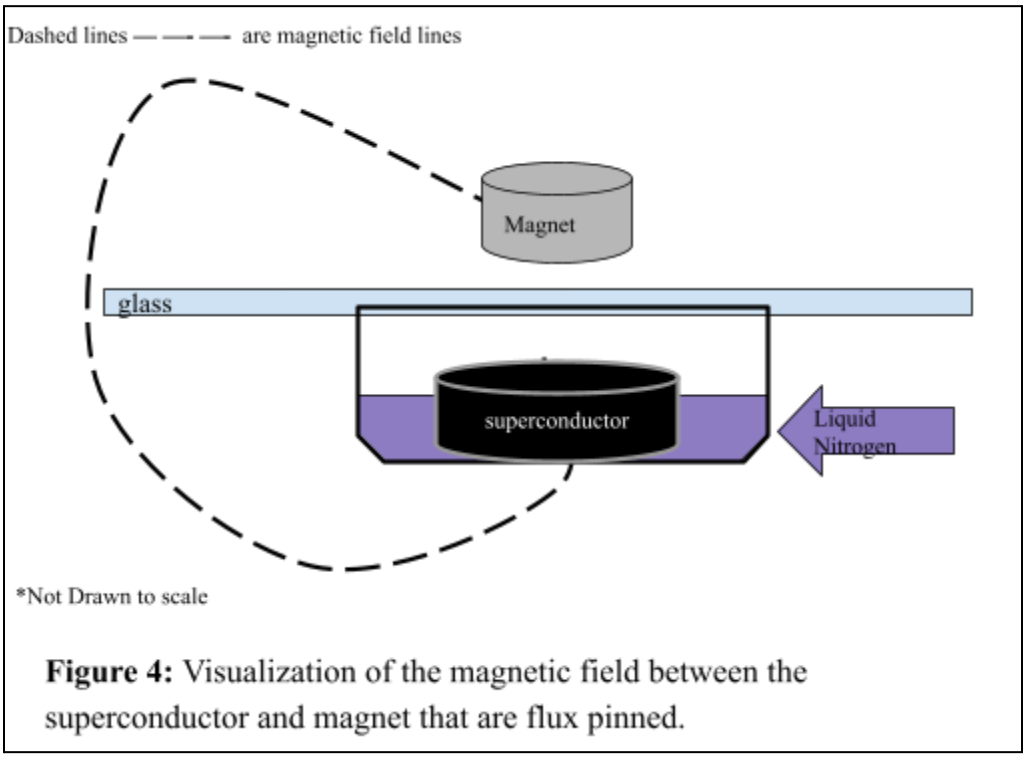


Figure 4: Visualization of the magnetic field between the superconductor and magnet that are flux pinned.

Materials List:

- 1 30 oz Thermos: capable of holding liquid nitrogen
- 28 oz of Liquid nitrogen
- Aluminum baking pan: 7.5 cm x7.5cm cm x 3.5 cm
- 1 Styrofoam block: 12.7cm x 12.7 cm x 12.7 cm
- 1 pair of boxcutter
- Pencil
- Ruler
- 9 Magnets: 1.2cm x1.2 cm x 0.6cm
- 1 Type II superconductor Yttrium barium copper oxide (Diameter: 14 mm, Height: 6 mm)
- 1 Piece of glass: 5cm x 14cm x 1mm
- 1 pair of antifreeze gloves
- 1 Timer

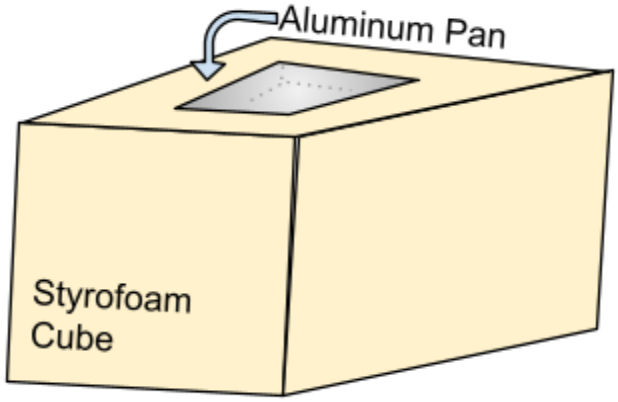


Figure 5: model of styrofoam container

Pre-lab:

1. Get the styrofoam container ready
    a. Use a pencil to trace an outline of the baking pan's surface onto the Styrofoam block.
    b. Measure a depth into the styrofoam block that matches the depth of the baking pan, and mark this part onto the block with the Pencil.
2. Cut the marked area out of the styrofoam block.
    a. Dispose of the small cube cut out of the styrofoam block.
    b. Place the Aluminum baking pan onto the incision made on the Styrofoam block.
3. Put on appropriate protective gear
    a. Put on clothing that fully covers all of one's body.
4. Measure the width, height, and length of the magnet being used in each trial using a ruler (in centimeters)

Lab

1. Place the superconductor inside the styrofoam container
2. Place glass on top of the aluminum baking pan, hovering over the superconductor
3. Place magnet on top of glass, near superconductor (1 to 2 inches)
4. Put on Anti-freeze gloves
5. Open thermos
6. Pour liquid nitrogen into the styrofoam container until it covers the superconductor
7. Close thermos
8. Wait (30 seconds)
9. Slowly Remove the glass pulling it away from the side and instantly start the timer
10. Stop the timer when the magnet hits the dish.

Post lab

1. Fill out the information chart :
    a. Independent: Surface area of the magnet
    b. Dependent: Time the magnet levitates
2. Properly dispose of other liquid nitrogen
3. Wait until the superconductor is at room temperature to start the next trial.

Revised Procedure:

Materials List:

- 1 30 oz Thermos: capable of holding liquid nitrogen
- 28 oz of Liquid nitrogen
- Styrofoam container: radius 3.25 cm x 1.5 cm
- 2 Styrofoam containers 3 cm in height
- 6 Magnets: 1cmx1cmx 0.3cm
- 1 Type II superconductor Yttrium barium copper oxide (Diameter: 14 mm, Height: 6 mm)
- 1 Piece of glass: 5cm x 14cm x 1mm
- 1 pair of antifreeze gloves
- Camera; 4k: 60 resolution

Lab

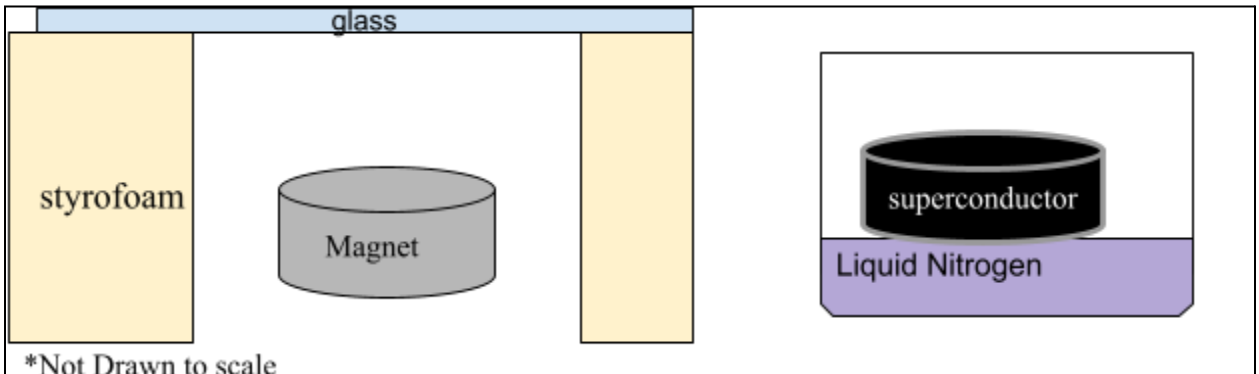


Figure 6: diagram of procedure

Pre-procedure:

1. Cut styrofoam cup to a small container of a 1.5 cm in height
2. Place the magnet/magnets you are using on the table or flat surface you have (with the large surface area acting as the base).
3. Place the 3 cm high styrofoam blocks beside the magnets and balance glass on top
4. Set up a camera to film the procedure, and start filming

Procedure

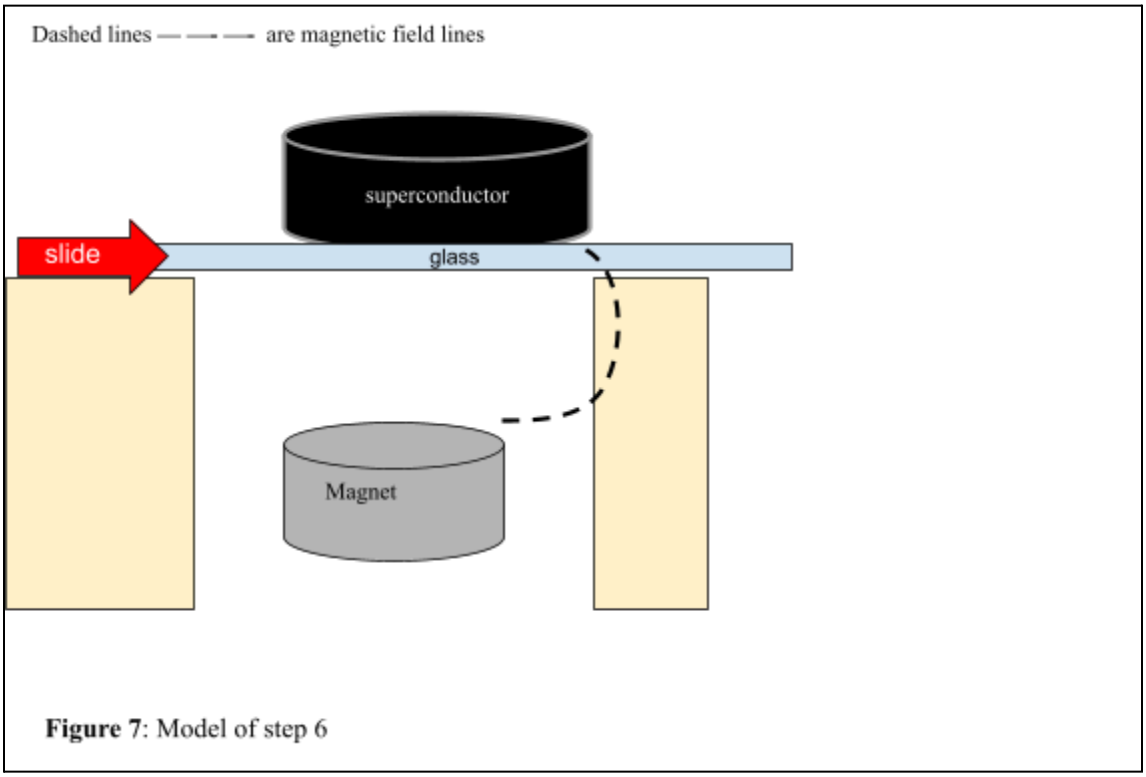


Figure 7: Model of step 6

1. Place the superconductor in a styrofoam cup
2. Pour liquid nitrogen into a styrofoam cup container until it covers the superconductor
3. Start the timer and wait for 30 seconds
4. Use tweezers to remove the superconductor from the styrofoam cup
5. Place the superconductor on the glass above the magnets.
6. Remove the glass by sliding it off carefully, attempting to not disturb the superconductor above.
7. Repeat steps 1-6, 7 times for each different-sized magnet, adding liquid nitrogen into the styrofoam cup as you see needed.

Post lab

1. Stop recording
2. Properly dispose of other liquid nitrogen
3. Fill out the information chart by using the *Seconds Count* app to break down the video's frames.

Explanation for revised lab procedure



Figure 8: Screenshot of Flux- Pinning Demonstration Video which the first lab procedure was based off of (Shore)

Change in Styrofoam Container

The first procedure included the utilization of an insulated apparatus designed to house liquid nitrogen and a superconductor. This container was constructed using a styrofoam cube and an aluminum pan. An aluminum pan was chosen because of its relatively low specific heat capacity. This property is advantageous for applications where rapid temperature changes are required (Short), particularly as the process of cooling a superconductor from room temperature to its critical temperature. The incorporation of styrofoam acted as an insulator, keeping the aluminum container at a stable temperature. It was assumed that this would prevent the excessive evaporation of liquid nitrogen before the superconductor could reach its critical temperature. However, the effects of the pan's surface area was not considered to be a problem within the initial procedure. The rate of heat transfer between the environment and the liquid nitrogen would be directly proportional to the surface area through which the heat is being applied. As a result, my first attempt towards the experiment was unsuccessful as the superconductor could not attain the critical temperature due to the liquid nitrogen evaporating too quickly. Furthermore, the change of container into one with a smaller surface area, yet still with the insulative properties, partly solved this problem.

Reason for no measurements of liquid nitrogen

At first I attempted to measure the liquid nitrogen poured into the styrofoam container. I soon realized that transferring the liquid nitrogen from the thermos to a measuring cup, and then to the container, took a significant amount of time and therefore a significant amount of liquid nitrogen to evaporate before it was actually used. The measuring of the liquid nitrogen with a measuring cup was significantly hard because of the smoke around the substance, allowing for a large margin of error.

Change in Glass used

The constant height of the glass over the liquid nitrogen container facilitated the balancing process between the superconductor and magnet. However, during the first trials of the 3rd experiment attempt, there were inconsistent results, with the flux pinning only occurring. I realized that the magnetic field was not strong enough to go through the initial glass used after. This is why the material list changed, the glass depth became slimmer.

Change in the Magents used

During my initial trials of my third attempt at this experiment I experienced the repulsion of the superconductor. This means that the superconductor always moved away from the magnet as I placed it above the magnet. This happened for the magnets that measured 1x1, 1x2, and 1x3. After observing this pattern I decided to focus my data on the magnets that had a bigger area than these. This mistake within trials reflected how I disregarded the Meissner effect. The lack of surface area meant that there would be fewer flux tubes available for the attraction of the superconductor and magnets. The lack of attraction meant that equilibrium would not be reached as the repulsion from the superconductor overcame the “strength” of the magnetic field.

Change in Superconductor and Magnet Position

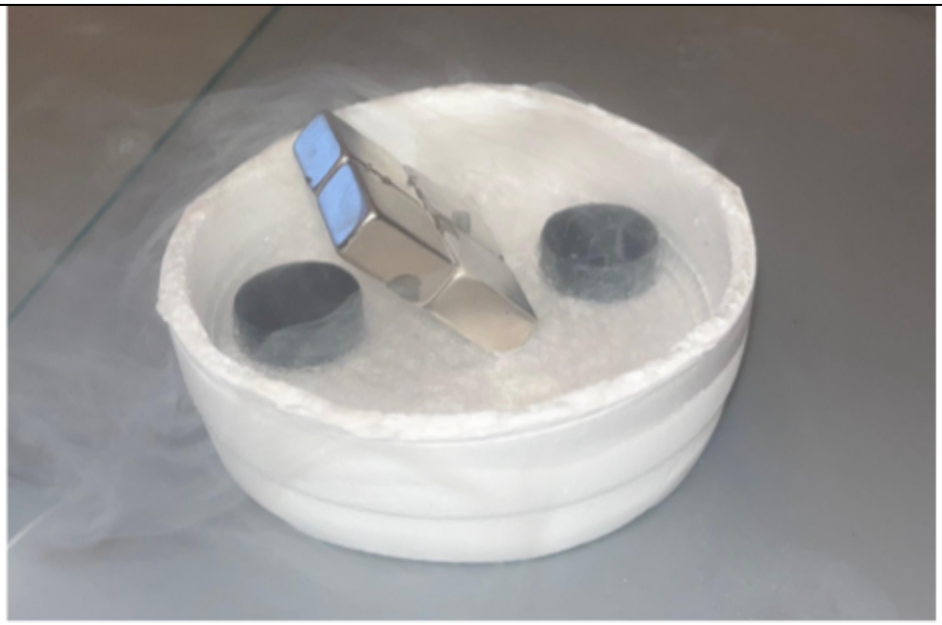


Figure 9: Magnet balancing between superconductors during second attempt of experiment

While the first drafts of my procedure instructed me to place the magnets above the cooled superconductor, the many failed trials pushed me to attempt other positions. While at first, I placed the superconductors below that it could keep its critical temperature – remaining in a liquid nitrogen solution– the best of my attempts resulted in the magnets semi-floating, balancing themselves between the two superconductors to be able to work. The reason behind these results was the mass of my magnets compared to my superconductors. Due to the fact that the Meissner effect is what allows for levitation, the mass difference between these objects made the gravitational force experienced by the Magnet too strong for the magnetic field force to combat. This resulted in them never reaching equilibrium and therefore no "floating" effect. By placing the object with the smaller mass – the superconductor– above the magnet, the magnetic field in the superconductor created a magnetic force that was able to balance out the magnitude of the force of gravity it experienced. While this change allowed for the acquisition of data, it made the results reflect the superconductor's change in temperature as it would now change its temperature at an accelerated rate to reach thermal equilibrium with the air around it.

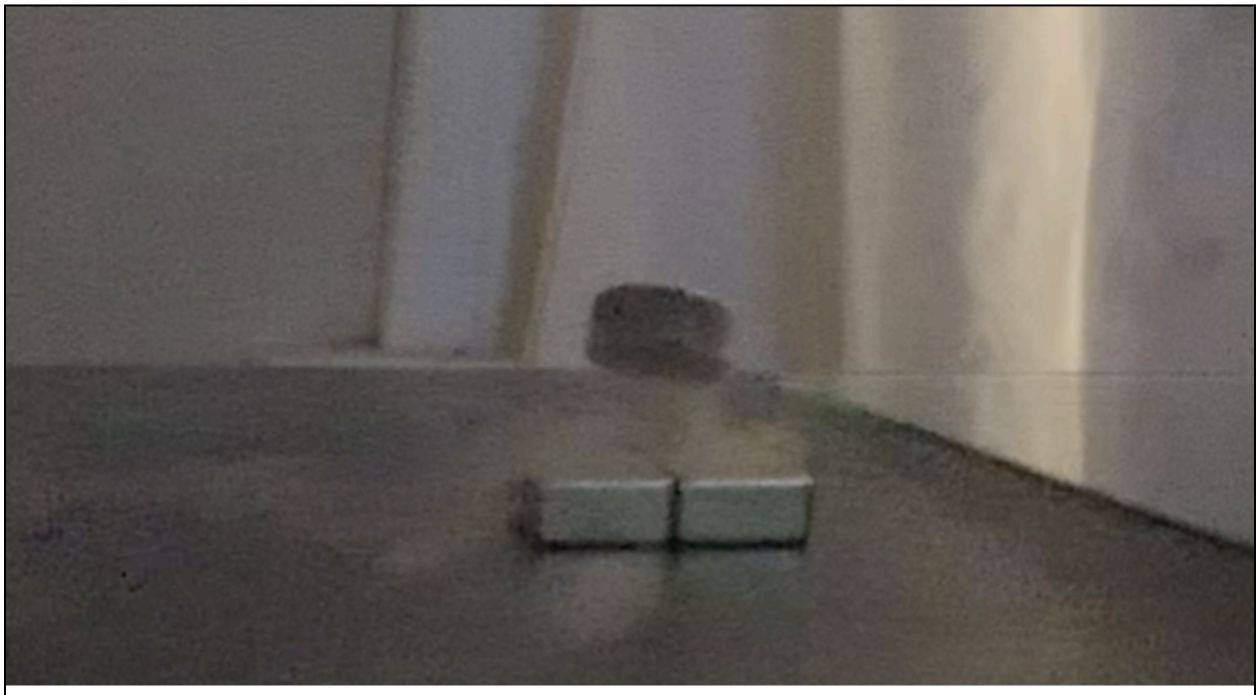


Figure 10: Frame from Quantum locking trial 3 2x2

Variables

Independent Variable — Dependent Variable

➢ Area (cm) — ➢ Time (sec)

Controlled Variables

Variable controlled	Why it's important to control	How this experiment will control or monitor the variable
Distance (cm) at which superconductor is placed above the magnet	According to Coulomb's law there is an inverse square relationship between distance and magnetic force. This makes the distance between magnets important. This is relevant within my experiment since if the distance of the superconductor and the magnet is too great the magnetic force is too weak. Therefore, the items would never achieve quantum locking because they would have no interaction. Similarly, if the distance from the superconductor to the magnet were minimal, the quantum locking may not be visible, with the items not having a gap that is prominent enough. In addition to this, since the dependent variable measures the time it takes the pined superconductor to reach the ground, having the same distance to travel for each trial would result in a common traveling time for each trial.	This experiment used a thin glass, placed at the same height for each trial. This allowed for the superconductor to be placed above it, stabilize itself, and "connect" to the magnet. The thin nature of the glass allowed for the magnetic field to pass through it with a similar strength and therefore pin the superconductor and magnet together. To remove the glass in a way that did not affect this displacement, I slid the glass using the styrofoam columns to guide this movement.
Time (sec) superconductor is placed in liquid nitrogen	This would ensure that the superconductor would reach its critical temperature and therefore have the properties to allow for quantum	Used a stopwatch to measure the amount of time that the superconductor was placed in the liquid nitrogen. Soaked in liquid

	locking.	nitrogen for at least the minimum about of time; 1 minute (“Superconductivity”).
Camera angle (rad)	Having the same camera angle is important because it would allow for the frames to observe all the trials from the same point of view, this would make the data precise. This allowed for the superconductor's movement to be interpreted in the same way.	Balanced the camera at 90 degrees.

Data

Raw Data

Area 4	Time levetating (±0.001 s)	Area 5	Time levetating (±0.001 s)	Area 6	Time levetating (±0.001 s)	Area 9	Time levetating (±0.001 s)
Trial 1	3.46	Trial 1	6.106	Trial 1	6.61	Trial 1	2.909
Trial 2	4.85	Trial 2	1.957	Trial 2	5.138	Trial 2	2.712
Trial 3	5.591	Trial 3	5.935	Trial 3	2.909	Trial 3	4.887
Trial 4	4.135	Trial 4	5.449	Trial 4	2.712		
Trial 5	5.835	Trial 5	4.199	Trial 5	4.887		
		Trial 6	3.003	Trial 6	4.141		
		Trial 7	2.15	Trial 7	4.569		

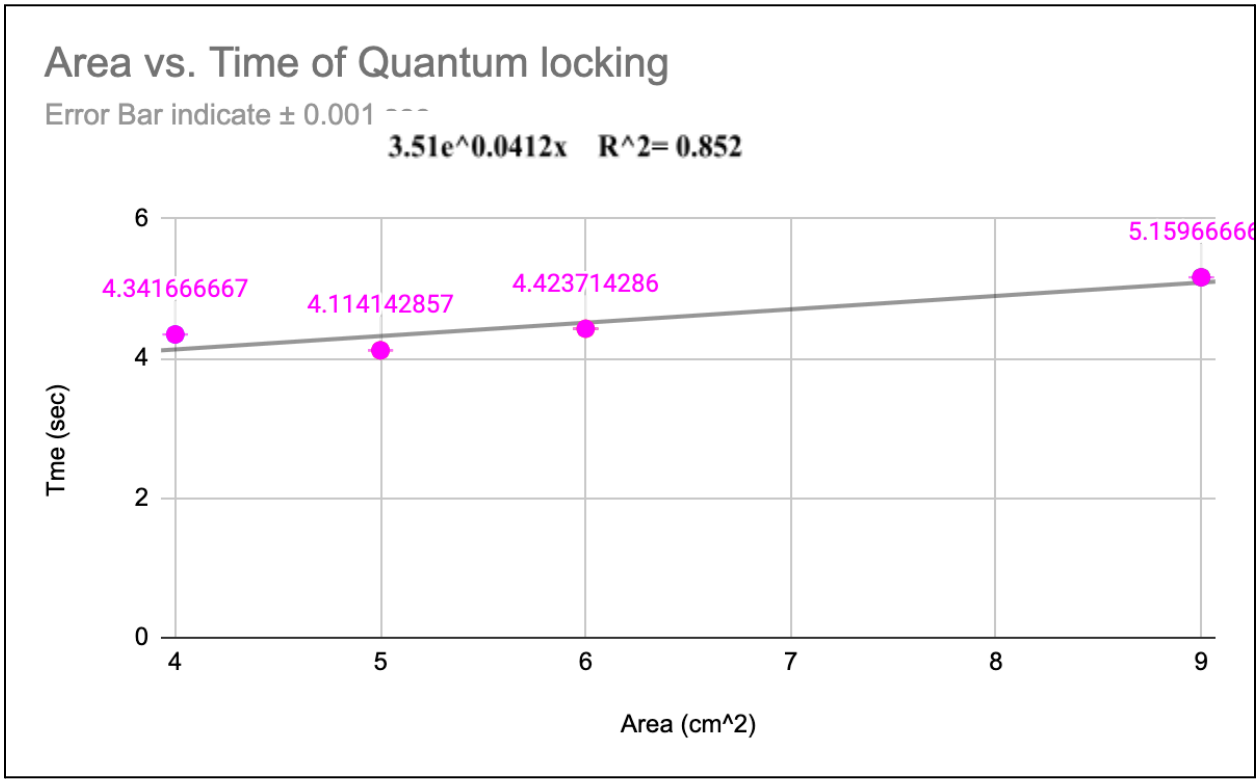
Data chart 1 : Measured time of levitation for each trial and magnet

Mean Value for Data

Area	4	5	6	9
Average time of levitation	4.341666667	4.114142857	4.423714286	5.159666667

Data chart 2 : Average time of levitation fpr each differently sized magnet

Graph



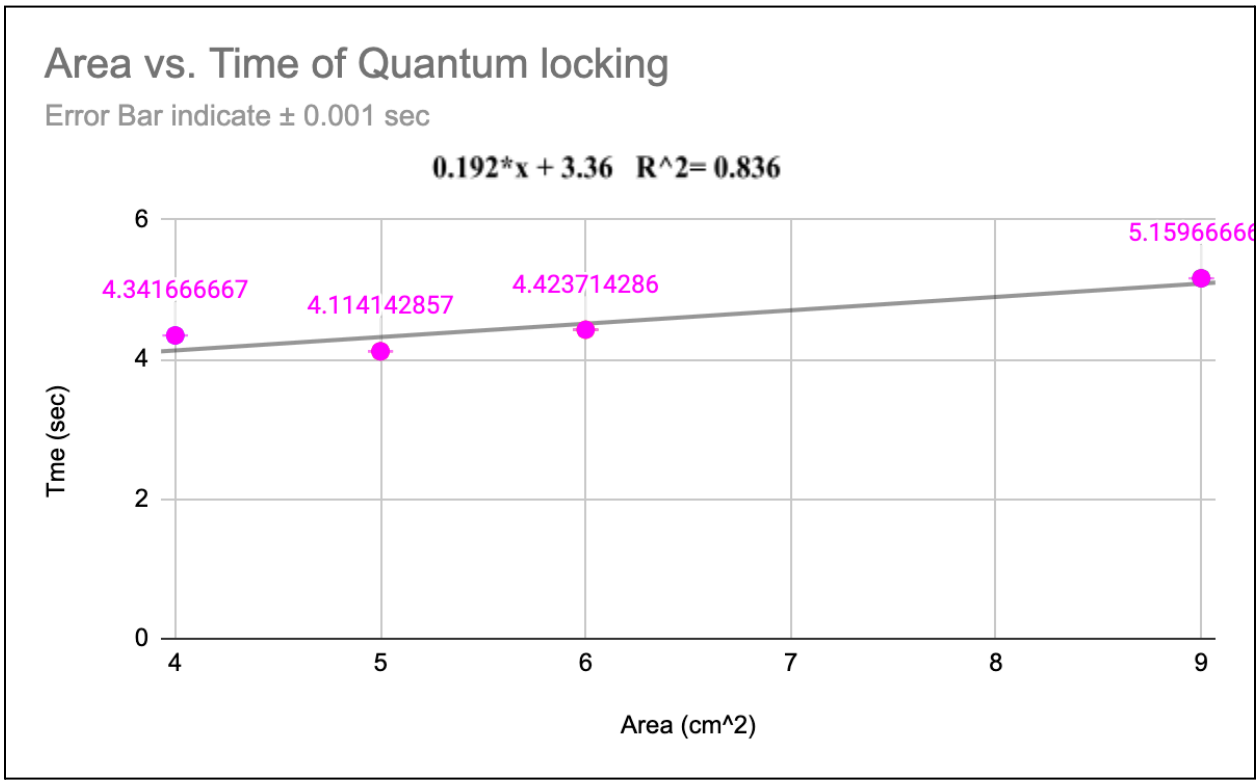
Analysis

It appears as if the greater a magnet's area is, the longer the superconductor levitates over it. This can be observed by the data I gathered and by the Magnetic flux equation. As we can tell by the relationship between magnetic flux – which allows for quantum locking— is proportional to the area that is perpendicular to the magnet. While the independent variable and dependent variable within this experiment were proportional, the observed data did not match the theoretical hypothesis as its relationship is not linearly proportional.

Data Analysis

When comparing my hypothesis to the data it appears that there were a lot of flaws within the data. This is especially prominent in the data gathered for the magnet with an area of 5 cm^2. This data point went against the hypothesis as the time that the superconductor levitated was less than the time it levitated for the area of 4 cm^2. This inconsistency begs to question of whether the levitation time is positively proportional to the area of a magnet. In addition, there was a large

range within the data, with the largest and smallest time intervals for levitation being 3.978 seconds apart. This is a considerable margin because of the small nature of the dependent value. These inconsistencies can be noted by the R^2 value of the logarithmic function of best fit. The 0.85 value represents how the line of best fit is only applicable to 85% of the data.



However, the logarithmic function was a close approximation of the relationship observed. This can be seen by the smaller R^2 of the linear function.

Impact of Error

This experiment was conducted during the summer, which means that the room temperature that was taken into account for my hypothesis, made my procedure flawed. 300 kelvins were assumed to be the surrounding environment's temperature when calculating the time needed for the superconductor to reach its critical temperature. This made the time I chose to submerge the superconductor insufficient, as the superconductor would not be at 77 kelvins within the 30 seconds it was submerged into the liquid nitrogen. While this might not have been obvious at first– due to the superconductor levitating– it is clear now that the superconductor did not match the Liquid nitrogen's temperature of 77 kelvin. While the superconductor was able to go below its critical temperature of 90 kelvin, it did not reach 77 kelvin which makes the data flawed as

this inconsistency makes the results a reflection of its temperature's rate of change rather than of the magnetic strength in relation to the magnet's area.
This error was especially prominent for the magnets that measured areas of four and nine. These magnets are missing data values because of the superconductor's inability to reach quantum locking.
In addition to not considering the actual temperature of the room. I did not consider the fluctuating temperature of the environment. Since this experiment was done in the afternoon, as time went by the temperature decreased. This would have resulted in trials progressively reflecting larger periods because the superconductor was changing temperatures at different rates as the experiment progressed.

Another source of error was the state at which the way I measured the time of levitation. Throughout the experiment, I measured the time of levitation starting with the moment I lost physical contact with the superconductor and ending with the moment that the superconductor made contact with the magnet (fell). This reflects a flaw within the experiment as it is possible that the superconductor was still levitating within one part, while the other stopped. This can be noted by how the superconductor descended, sometimes touching the magnet while a part still levitated.

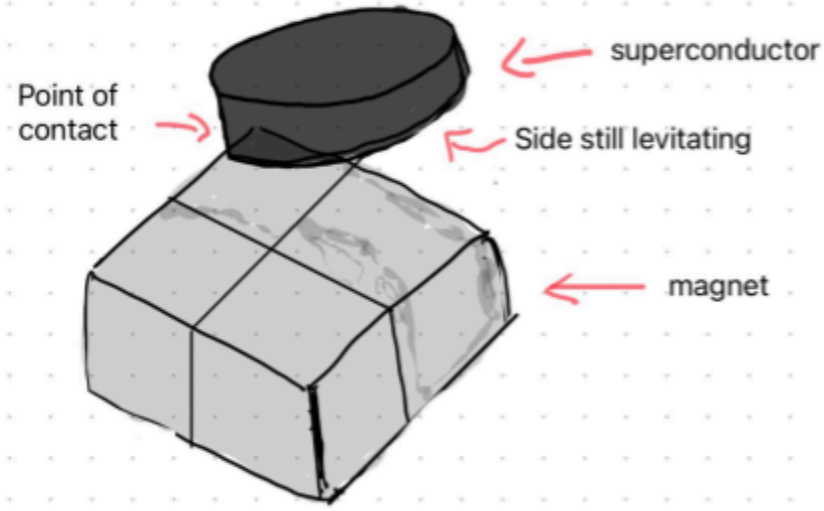


Figure 11: Visual of inconsistent fall of superconductor

Discussion

Now that I have addressed the significant errors within this experiment, I can address the improvements.

The most significant change I would like to make is how I interpreted the time of levitation. I would change how I defined when the superconductor officially stopped levitating. At times the superconductor would slat and touch the magnet below it while the other side was suspended in the air. While I interpreted this physical contact as the end of each trial to have a consistent procedure, obtaining the average time value for the first and second time of contact between the superconductors could have made my data more consistent.

Another limitation that this experiment had was its dependence on visual interpretations of the data. The liquid nitrogen made the superconductor emit vapor, blocking the camera's view and therefore the data which relied on observing the time at which the superconductor made contact with the magnet. I would change the experiment by using photogates to measure the period to make the data more precise.

To combat the fluctuating temperature of the environment, an improvement I would conduct this experiment in an environment in an air-conditioned room as this would result in a consistent temperature. Furthermore, I would use this consistent temperature to solve for the maximum time of levitation value using Newton's Law of Cooling.

Conclusion

This investigation used various magnets and a superconductor to explore the relation between the size of magnets and their effect on the quantum locking process. Will a bigger magnet allow for this position to last longer?

It appears that as the area of a magnet increases, type II superconductors can be suspended in the air for longer periods. This is a result of a stronger magnetic field. With this data there is a logarithmic proportionality between the duration of the superconductor's levitation and the area of the magnet it is suspended over.

Taking the uncertainty value into consideration leaves space for more research to confirm these findings and expand on them. However, this investigation can still apply to the current goal of revolutionizing technologies through quantum locking. Researching Type II superconductors that achieve a superconducting state at or above room temperature has been sought after to achieve this goal. Given that the only way to achieve this superconducting state under high temperatures is by applying extreme pressure on the superconductors, using stronger magnets offers another way to make superconductivity an applicable phenomenon in our daily lives. This application can be "the development of frictionless joints for energy conservation, and next-generation space systems for launching, docking, object manipulation, and satellite formations"("Full Abstract").

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